MIL-E-1/1762
16 Sept 1975
SUPERSEDING
MIL-E-1/1119H
23 February 1973

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

1/ TYPE 8979

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, integral cavity, mechanically tuned from 8,500 to 10,500 MHz, waveguide output.

ABSOLUTE RATINGS:

Parameter:	Ef	Ehk	Ers	Er	Ik	T(shell)	Alt
Unit:	V	Vdc	Vdc	Vdc	mAdc	°C	ft
Maximum:	6.8	±45	330	-500	37	110	10,000
Minimum:	5.8	---	---	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Weight:	8 ounces (maximum)
Base:	See figure 1	Mounting position:	Any
Cathode:	Unipotential	Installation:	See note 8
Mounting:	See figure 1		

TEST CONDITIONS:

Parameter:	EF	Ehk	Ers	Er	T(shell)	VSWR	F
Unit:	V	Vdc	Vdc	Vdc	°C	σ'	MHz
	6.3	0	300	-45 to -180	Note 1	1.1:1(max)	F2

FREQUENCY		
F	MHz	±
F1	8,500	15
F2	9,500	15
F3	10,500	15

GENERAL:

Qualification - Required

1/ Replaces Varian tube type VRX-7213A and Sperry tube type SRX-92. Supersedes tube type 7511 (MIL-E-1/1119H).

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
1042	Shock	2	Test condition J, except 15 G	ΔPo/Po	---	±10	%
4027	Temperature coefficient	-	Mode A; TA = 20° to 80°C	---	---	+0.3 -0.2	MHz/°C MHz/°C
4225	Residual FM	3	Er(mode A)/max Po	ΔF	---	80	kHz (p to p)
	Quality conformance inspection, part 1	4					
1256	Electrode current (cathode)	-	Er(mode A)/max Po	Ik	---	34	mAdc
4214	Cathode emission	-	Ef = 5.8 V; Er(mode A)/max Po	ΔIk/Ik	---	-15	%
4223	Mechanical tuning range	-	Er(mode A)/max Po;				
			Low frequency	F	---	8,500	MHz
			High frequency	F	10,500	---	MHz
4250	Power output	-	F = F1 to F3; Er(mode A)/max Po	Po	20	---	mW
4213	Reflector voltage	-	F1, F2, F3; Er(mode A)/max Po				
	(1)	-		Er	-45	-80	Vdc
	(2)	-		Er	-85	-135	Vdc
	(3)	-		Er	-130	-180	Vdc
4229	Total reflector current	5, 6	Er = -140 to -200 Vdc	Ir	---	10	μAdc
	Quality conformance inspection, part 2						
1031	High-frequency vibration	---	No voltages applied	---	---	---	---
1211	Insulation of electrodes	---	300 Vdc; tube cold	Ikrs	---	15	μAdc
				Ifrs	---	15	μAdc
4289	Heater current, nonoperating	---		If	---	600	mA
4270	Heater-cathode leakage	---	Ehk = +45 Vdc	Ihk	---	100	μAdc
4229	Reflector-leakage current	5, 6	Er = -140 to -200 Vdc	Ir	---	6	μAdc
4229	Reflector gas current	5, 6	Er = -140 to -200 Vdc	Ir	---	4	μAdc
4231	Electronic tuning hysteresis	7	F = F1 to F3; Er(mode A)/max Po	Ratio	---	0.5	---
4280	Electronic tuning range	-	F = F1 thru F3; Er(mode A)/50% max Po	ΔF	30	---	MHz
4215	Reflector voltage modulation coefficient	-	Method B; F = F1; Er(mode A)/max Po	ΔF/Er	2.5	4.0	MHz/v
4223	Mechanical tuning rate	-	F = F1 to F3	---	---	---	---
4223	Operating torque	-		---	---	2.5	in.-lb

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test	---	Group D	t	2,000	---	hrs
---	Life-test end point:						
4250	Power output	---	F = F1 to F3; Er(mode A)/max Po	Po	15	---	mW

NOTES:

1. Shell temperature shall be measured at the point indicated on figure 1.
2. The TUT shall be operating and ΔPo/Po shall be measured between successive shock pulses.
3. The peak-to-peak frequency deviation arising from the use of ac on the heater shall be less than the value specified herein for any power frequency from 50 to 1,200 Hz.
4. Unless otherwise specified, the AQL for quality conformance inspection, part 1, shall be 1.0, inspection level II.
5. This test shall be performed at the conclusion of the holding period.
6. The tube shall not be oscillating during the test.
7. Superimposing sufficient 60 Hz ac voltage on the direct-reflector voltage to suppress oscillations on the ends of the sweep, the mode curve shall be observed on an oscilloscope whose amplifier has minimum bandpass of 0.1 MHz.
8. Tubes procured to this tube specification sheet shall include complete installation instructions for replacing the superseded tube types.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force -
DSA - ES

User activities:
Army - MC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3001)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.

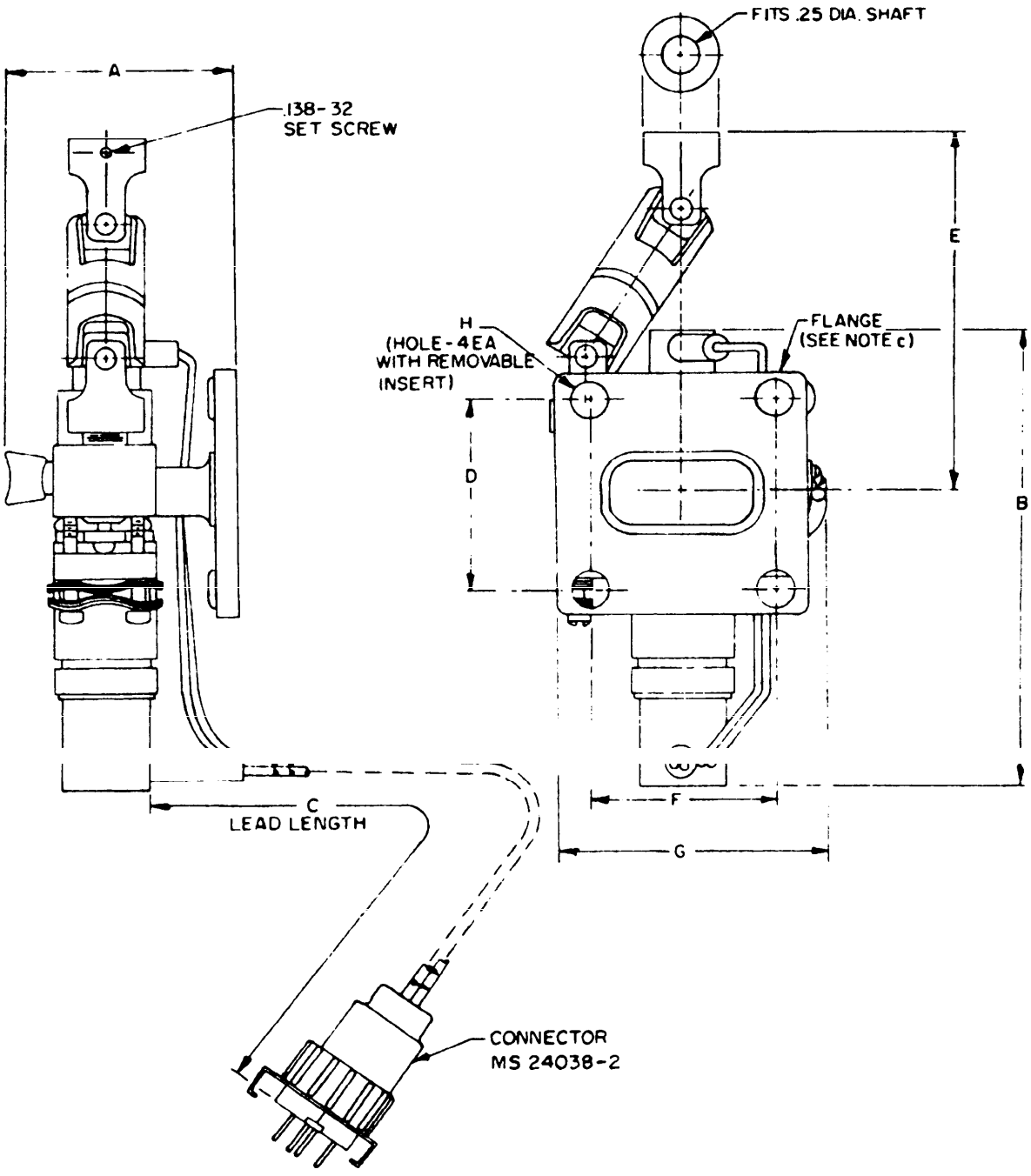


FIGURE 1. Outline drawing of electron tube type _____.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.

Ltr	Dimensions in inches with metric equivalent (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1 (See note 4)		
A	---	1.750 (44.45)
B	---	3.250
C	8.500	9.500
D	1.276	1.284
E	---	2.850
F	1.216	1.224
G	---	1.875
H	(.192 Diameter)	

Base connections		
A	Gray	Reflector
B	Brown	Body
D	Green	Cathode
E	Yellow	Heater
H	Yellow	Heater

NOTES:
a. Dimensions are in inches.
b. Metric equivalent (to the nearest .01 mm) is given for general information only and is based upon 1 inch = 25.4 mm.
c. Flange shall be designed to mate with flange UG-39/U of MIL-F-3922/1, or equivalent.
d. Orientation of pinch-off shall be at the manufacturer's option.

FIGURE 1. Outline drawing of electron tube type - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

Source: https://assist.dla.mil -- Downloaded: 202610-15T15:32Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.